

Robert Jenkinson Alvarez

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EDUCATION

London School of Economics and Political Science (LSE)

MSc Financial Mathematics (Distinction)

Sep 2023 – Jul 2024

- Black-Scholes; Interest Rate/Credit Risk-Hull-White/Vasicek/HJM; Stochastic Processes-Ito; Fixed Income; Portfolio Management.
- Developed finite-difference PDE solvers, variance-reduced Monte Carlo and Euler SDE schemes, analysing stability/convergence.
- Quant Algathon (Top 2): Developed a 3-layer LSTM forecasting BTC/ETH returns using sentiment data, with 1.2 out-of-sample Sharpe and 54% directional accuracy.

University of Bath

BSc Mathematics (First-Class Honours)

Sep 2020 – Jul 2023

- Probability: Markov Processes; Martingales; Stochastic Systems; Discrete Prob.; Machine Learning; Measure Theory; Statistics.
- Pure/Applied: Advanced Real/ Complex Analysis; Algebra; ODEs/Control Theory; PDEs/Vector Calculus; Numerical Analysis (97%).

RESEARCH & ENGINEERING EXPERIENCE

Machine Learning Researcher

Beyond Isotropy in JEPAs: HamJEPA (arXiv:2605.20107); NeurIPS 2026 under review

Oct 2025 – Present

- Established theory for structured priors in JEPAs, proving that under an H-energy budget the optimal embedding prior is a canonical Hamiltonian-Gaussian via minimax/maximum-entropy and deriving a closed-form price of isotropy bound.
- Introduced a phase-space JEPA encoding views as content/momentum coordinates predicting transitions with a vol-preserving, time-reversible symplectic leapfrog integrator, replacing isotropic Gaussian regularisation with structured dynamical priors.
- Improved LeJEPA/SIGReg top-1 (linear-probe/kNN-20) on CIFAR-100 by +10.6/+6.5 (80 epochs) and +3.52/+4.89 (30 epochs) and on ImageNet-100 +7.52/+4.82 (45 epochs), raising effective-rank proxy from 109.7 to 125.6 and 51.2 to 96.5 respectively.
- Built a config-driven PyTorch training stack featuring differentiable Hamiltonian-flow prediction, separable Hamiltonians, symplectic Euler and velocity-Verlet integrators, mixed-precision-safe fp32 integration and optional bidirectional consistency.

Mathematics, Finance & Machine Learning Research

Variational Rigidity of Minimising Foliations

Jan 2026 – Present

- Proved the one-dimensional and arbitrary-dimensional cases of a stated open rigidity problem concerning minimising foliations of periodic semilinear elliptic equations.
- Innovated a combination of global foliation coordinates, variational dynamics and Hopf-Riccati rigidity, whereas the higher dimensional argument combines calibrated variational identities, Runge approximation and Bloch/Fermi spectral separation.

Geometric Stochastic Modelling & Manifold Optimization

Sep 2024 – Present

- Architected an end-to-end world-model platform for continuous, high-dimensional market data, fusing spot, rates and option surfaces into geometry-aware states across eight coupled assets with 3,266 certified states and zero failures.
- Designed a foliation-structured HamJEPA world model that organises regimes as latent leaves, learns within-leaf evolution through self-supervised symplectic prediction and uses transverse motion as an explicit regime-change coordinate.
- Converted latent-state transitions into cross-asset forecasts, tested out of sample under costs and negative controls.

Arbitrage-free Option surfaces via Chebyshev Tensors and Hamiltonian Fog (arXiv:2512.01967)

Aug 2025 – Nov 2025

- Designed an arbitrage-free equity option surface for liquid options via a Chebyshev tensor basis in moneyness and maturity space.
- Formulated calibration as a large-scale convex QP, encoding no-arbitrage constraints as sparse linear operators on a collocation grid, with orthogonal projections achieving 98-99% inside-spread and <1% static no-arb violations in calm regimes.
- Pioneered a fog layer modelling noise as a 3D density, minimising a convex energy with graph-Laplacian smoothness and band potentials, relaxing conflicting quotes while keeping the global surface arbitrage-free with <1% violations.

Optiver

Derivatives Quant Trading & Market Making Trainee

Oct 2023 – Nov 2023

- Developed Black-Scholes based options market-making algorithms in a 3-person team, optimising bid-ask spreads by dynamically weighting liquidity tiers and order book depth in Optibook's live trading simulated exchange.
- Reduced daily PnL variance by 30% and 95% VaR by 25% via delta-hedging and gamma rebalancing (IOC orders) over 4 weeks.
- Improved order placement with a priority queue architecture, slashing latency by 40% through pruning low-liquidity strike/expiry pairs and caching Greeks calculations in real time for high-frequency updates.

SKILLS & INTERESTS

- **Technical Skills:** Python (NumPy, Pandas, TensorFlow, SciPy, QuantLib, OSQP, PyTorch), MATLAB, R, SQL, LaTeX.
- **Communication:** Tutored metric spaces & stochastic processes to 5 university students, improving pass rates by 10%.
- **Methods:** self-supervised learning, numerical optimisation, time-series, convex programming, Monte Carlo, PDE/SDE methods.
- **Languages / Interests:** Native Spanish & English (CPE C2). Piano (ABRSM Grade 8), Giveaway chess (2220 Elo, top 100 worldwide).